2023-24

S6 MATH CP

PAPER 1

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Six First Term Examination

MATHEMATICS Compulsory Part

PAPER 1

Question-Answer Book

06/11/2023

8:15 a.m – 10:30 a.m (2¼ hours)

This paper must be answered in English

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.

2. This paper consists of THREE sections, A(1), A(2) and B.

3. Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

4. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string INSIDE this book.

5. Unless otherwise specified, all working must be clearly shown.

6. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.

7. The diagrams in this paper are not necessarily drawn to scale.

8. No extra time will be given to candidates for writing your Name, Class and Class number or filling in the question number boxes after 'Time is up' announcement.

Name: _____

Class: _____( )

	Marker Use Only
Question	Mark
1	
2	
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Total			

SECTION A(1) (35 marks)

1. Simplify $\frac{(a^{-4}b)^2}{a^5}$ and express your answer with positive indices. (3 marks)

2. Factorize
   (a) $9x^2 - 30xy + 25y^2$,
   (b) $9x^2 - 30xy + 25y^2 - 3x + 5y$.

(3 marks)

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3. (a) Round up 160.725 to 1 decimal place.
   (b) Round off 160.725 to 2 significant figures.
   (c) Round down 160.725 to the nearest integer.

(3 marks)

4. In a summer camp, the ratio of the number of boys to the number of girls is 8 : 7. If 16 boys and 11 girls leave the summer camp, then the number of boys and the number of girls are the same. Find the original number of girls in the summer camp.

(4 marks)

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5. The marked price of a video camera is 20% higher than the cost. If the video camera is sold at a 20% discount on the marked price, does the seller suffer a loss? Explain your answer. (4 marks)

6. (a) Solve the inequality $\frac{4-3x}{5} \leq 1-x$.

(b) Sarah claims that there are no integers satisfying both the inequalities $\frac{4-3x}{5} \leq 1-x$ and $2-x<3$. Do you agree? Explain your answer.

(4 marks)

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7. The following table shows the numbers of pens in the pencil boxes of a group of students.

Number of pens	5	6	7	8	9
Number of students	42	8	x	20	15

It is known that the median number of pens is 6.5.

(a) Find the value of x.

(b) Find the mean and the standard deviation of the numbers of pens.

(4 marks)

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8. In Figure 1, AB // CD. E is a point lying on AD such that $AC = AE$. Find x, y and z.

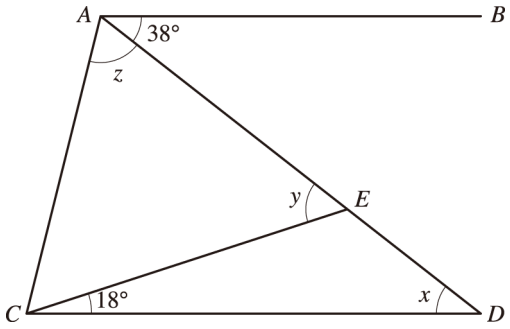


Figure 1

(5 marks)

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9. In a polar coordinate system, O is the pole. The polar coordinates of the points P and Q are $(k, 34°)$ and $(10, 304°)$ respectively, where k is a positive constant. It is given that $PQ = 26$.

(a) Is ΔOPQ a right-angled triangle? Explain your answer.

(b) Find the perimeter of ΔOPQ.

(5 marks)

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SECTIONS A(2) (35 marks)

10. Figure 2(a) shows a right circular conical jelly of base radius 10 cm and height 3.87 cm and a hemispherical jelly of radius R cm. The conical jelly and the hemispherical jelly are melted and recast into the lower part (I) and upper part (II) of a jelly as shown in Figure 2(b) respectively. The jelly formed is in the form of a frustum which is made by cutting off the lower part of an inverted right circular cone of base radius r cm and height 10 cm. The height of the jelly formed is 5 cm and the height of the lower part of the jelly is 3 cm.

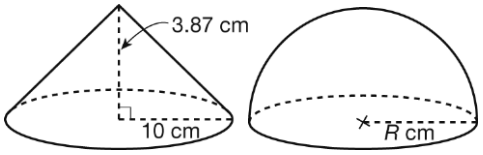


Figure 2(a)

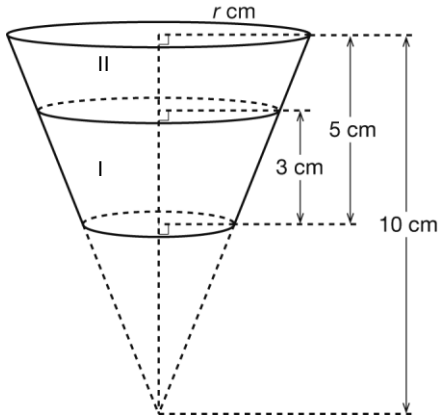


Figure 2(b)

(a) Find the volume of the conical jelly. (Give the answer in terms of π.) (1 mark)

(b) Find r and R. (5 marks)

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11. It is given that y is the sum of two parts. One part varies directly as x^4 and the other part varies directly as x^2. When $x = 2$, $y = 4$ and when $x = 4$, $y = 208$.

(a) Express y in terms of x. (3 marks)

(b) Find the value(s) of x when $y = 54$. (4 marks)

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12. Let $f(x) = (x + m)(x + n)(x + 3) + 1$, where m and n are positive integers and $m < n$. It is given that $f(x)$ is divisible by $x + 2$.

(a) (i) Find the values of m and n.

(ii) Factorize $f(x)$.

(5 marks)

(b) Solve the equation $f(x) = x^2 + 3x + 2$. (2 marks)

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13. In Figure 3, O is the centre of the circle $ABCD$. AB is a diameter of the circle. AC cuts OD and BD at E and F respectively. It is given that $\Delta AEO \sim \Delta DEF$.

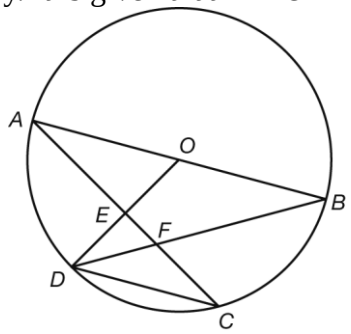


Figure 3

(a) Daisy claims that BD is the angle bisector of $\angle ODC$. Do you agree? Explain your answer. (2 marks)

(b) Prove that AB // DC. (2 marks)

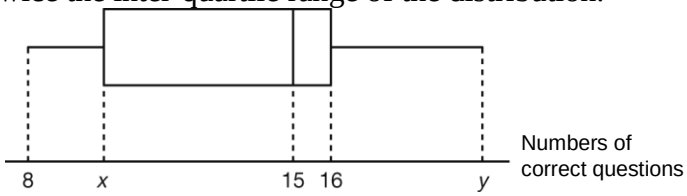
(c) If E is the mid-point of AC, is F the mid-point of EC? Explain your answer. (3 marks)

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14. The box-and-whisker diagram below shows the distribution of the number of correct questions of 12 students in a Mathematics competition. It is known that the range of the distribution is twice the inter-quartile range of the distribution.



(a) How many possible pairs of values of x and y are there? (4 marks)

(b) After review, the marker found that the numbers of correct questions of 4 students were recorded wrongly. The table shows the original and amended numbers of correct questions of those 4 students.

Student	Original numbers of correct questions	Amended numbers of correct questions
A	8	7
B	8	9
C	18	21
D	20	26

It is given that the mean of the original numbers of correct questions in the distribution is 14. Find the mean, median and range of the amended numbers of correct questions. (4 marks)

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SECTION B (35 marks)

15. The standard deviation of the scores of a class of students in a test is 2.5 marks. The scores of Roy and Cheryl in the test are 42 marks and 70 marks respectively. The standard score of Roy in the test is –4.8.

(a) Find the standard score of Cheryl in the test.

(2 marks)

(b) A student, Bosco, withdraws from the class and his test score is then deleted. It is given that his score is 54 marks. Will there be any change in the standard score of Cheryl due to the deletion of the test score of Bosco? Explain your answer.

(2 marks)

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16. A committee consists of 6 male members and 8 female members. Now 4 members are selected from the committee at random.
    (a) Find the probability that 2 male and 2 female members are selected. (3 marks)
    (b) Find the probability that the numbers of male and female members selected are different. (2 marks)

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17. Figure 4 shows a regular tetrahedron *ABCD*. *M* is the midpoint of *BC*.

(a) Find *AM*. (2 marks)

(b) Find the angle between the planes *ABC* and *BCD*. (3 marks)

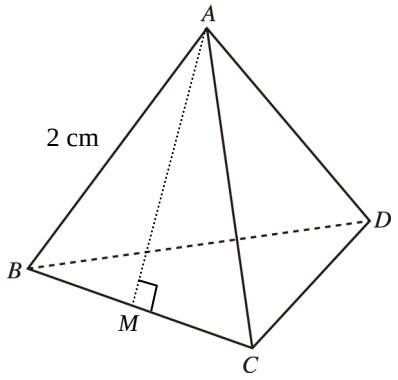


Figure 4

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18. (a) In Figure 5, the equation of the straight line L_1 is $2x + y = 8$ and the y-intercept of the straight line L_2 is 4. L_1 and L_2 intersect at (3, 2). The shaded region (including the boundary) represents the solution of a system of inequalities. Find the system of inequalities. (4 marks)

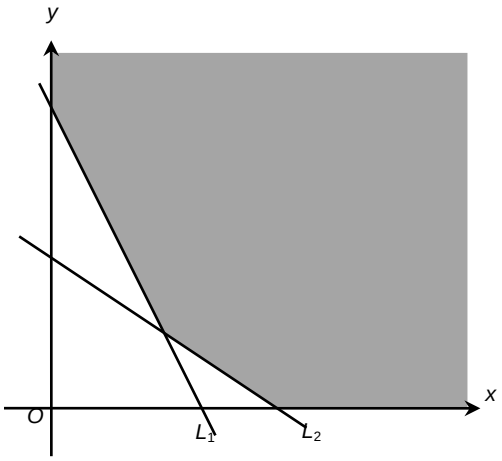


Figure 5

(b) Natalie wants to buy some greeting cards and stickers as a gift. There are two different packages *A* and *B* in a stationery shop. Each package contains greeting cards and stickers. The numbers of greeting cards and stickers in each pack of packages *A* and *B* and their prices are given below:

	Greeting card	Sticker	Price ($)
Package *A*	10	18	15
Package *B*	15	9	12

Natalie wants to buy at least 60 greeting cards and 72 stickers. Suppose that she buys x packs of package *A* and y packs of package *B*. Is it possible that Natalie buys the gift with $70? Explain your answer.

(4 marks)

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19. Ryan and Gordon play a game consisting of two rounds. In the first round, Ryan and Gordon take turns to throw a fair die. The player who first gets 1 or 6 wins the first round. Ryan and Gordon play the first round until one of them wins. Ryan throws the die first.

(a) Find the probability that Gordon wins the first round of the game.

(3 marks)

(b) In the second round of the game, cards are drawn one by one randomly from 9 cards numbered with 1, 2, 3, …, 9 respectively.
The player of this round adopts one of the following two options.

Option 1: The player draws two cards with replacement. If the two cards drawn are with the same number, the player wins a $20 coupon. If the two cards drawn are with two consecutive numbers, the player wins a $10 coupon. Otherwise, the player wins nothing.

Option 2: The player draws three cards with replacement. If the three cards drawn are with the same number, the player wins a $100 coupon. If the three cards drawn are with three consecutive numbers, the player wins a $20 coupon. If two of the three cards drawn are with two consecutive numbers and the remaining card is the same as one of the other two cards, the player wins a $10 coupon. Otherwise, the player wins nothing.

(i) If the player of the second round adopts Option 1, find the expected amount of coupon won.

(ii) Which option should the player of the second round adopt in order to maximize the expected amount of coupon won? Explain your answer.

(iii) Only the winner of the first round plays the second round. It is known that Ryan will adopt Option 2 and Gordon will adopt Option 1 in the second round. Who, Ryan or Gordon, is more likely winning nothing in the game? Explain your answer.

(10 marks)

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END OF PAPER

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